B. JoJo's Incredible Adventures

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: math, strings, two pointers | https://codeforces.com/problemset/problem/1820/B

Did you think there was going to be a JoJo legend here? But no, that was me, Dio!

Given a binary string s of length n, consisting of characters `0` and `1`. Let's build a **square** table of size $n \times n$, consisting of `0` and `1` characters as follows.

In the first row of the table write the original string s. In the second row of the table write *cyclic shift* of the string s by one to the right. In the third row of the table, write the cyclic shift of line s by two to the right. And so on. Thus, the row with number k will contain a cyclic shift of string s by k to the right. The rows **are numbered from 0 to n - 1 top-to-bottom**.

In the resulting table we need to find the rectangle consisting only of ones that has the largest area.

We call a rectangle the set of all cells (i, j) in the table, such that $x_1 \le i \le x_2$ and $y_1 \le j \le y_2$ for some integers $0 \le x_1 \le x_2$ lt n and $0 \le y_1 \le y_2$ lt n.

Recall that the cyclic shift of string s by k to the right is the string $s_{n-k+1} \dots s_n s_1 s_2 \dots s_{n-k}$. For example, the cyclic shift of the string "`01011`" by 0 to the right is the string itself "`01011`", its cyclic shift by 3 to the right is the string "`01101`".

Input

Each test consists of multiple test cases. The first line contains a single integer t ($1 \le t \le 2 \cdot 10^4$) — the number of test cases. The description of test cases follows.

The first and the only line of each test case contains a single binary string s ($1 \le |s| \le 2 \cdot 10^5$), consisting of characters `0` and `1`.

It is guaranteed that the sum of string lengths $|s|$ over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the maximum area of a rectangle consisting only of ones. If there is no such rectangle, output 0.

Example

Input
5 0 1 101 011110 101010

Output

```
0
1
2
6
1
```

Note

In the first test case, there is a table 1 × 1 consisting of a single character `0`, so there are no rectangles consisting of ones, and the answer is 0.

In the second test case, there is a table 1 × 1, consisting of a single character `1`, so the answer is 1.

In the third test case, there is a table:

1	0	1
1	1	0
0	1	1

In the fourth test case, there is a table:

0	1	1	1	1	0
0	0	1	**1**	**1**	**1**
1	0	0	**1**	**1**	**1**
1	1	0	0	1	1
1	1	1	0	0	1
1	1	1	1	0	0

In the fifth test case, there is a table:

1	0	1	0	1	0
0	1	0	1	0	1
1	0	1	0	1	0
0	1	0	**1**	0	1
1	0	1	0	1	0
0	1	0	1	0	1

Rectangles with maximum area are shown in bold.